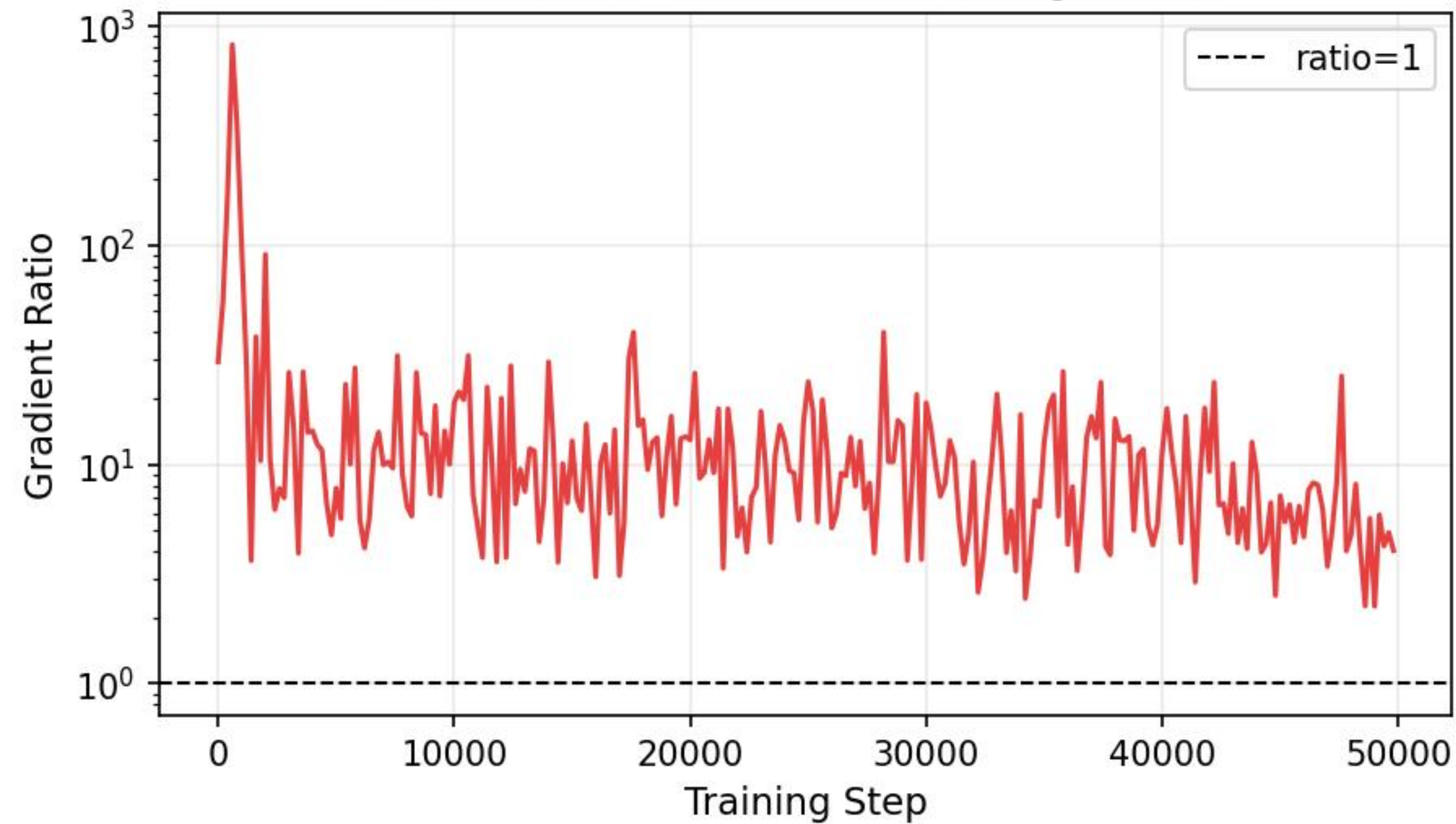


Gradient Magnitude Ratio $|\nabla L_{\text{pde}}|/|\nabla L_{\text{bc}}|$ ($\beta=30$)

Gradient Ratio Over Training
(sustained >1 = PDE dominates IC/BC gradients)



Loss Component Trajectories

